

Chapter 5 — Sequences & Mathematical Induction

Part 2: 5.2 & 5.3 Mathematical Induction

COMP 233 Discrete Mathematics | Birzeit University
Content follows Epp, Discrete Mathematics with Applications (4th ed.).

5.2 & 5.3 Induction - in this lecture

- 1 **What induction is, and why it works**
- 2 The method, and a first full proof
- 3 Sums: Gauss, arithmetic and geometric formulas
- 4 Divisibility and inequalities
- 5 Properties of recursive sequences
- 6 Induction vs. deduction, and more practice

A puzzle that needs a new kind of proof

The question

Using only 3-cent and 5-cent coins, which total amounts n can you pay exactly?

$$8 = 3 + 5, \quad 9 = 3 + 3 + 3, \quad 10 = 5 + 5, \quad 11 = 3 + 3 + 5$$

The pattern, and the problem

Every amount from 8 upward seems to work - but "8, 9, 10, 11, ... and onward forever" is infinitely many cases.

We cannot check them one by one. We need a method that proves a statement for all integers $n \geq 8$ at once. That method is mathematical induction.

The principle of mathematical induction

To prove $P(n)$ is true for all integers $n \geq a$

Basis step

(1) Basis: $P(a)$ is true

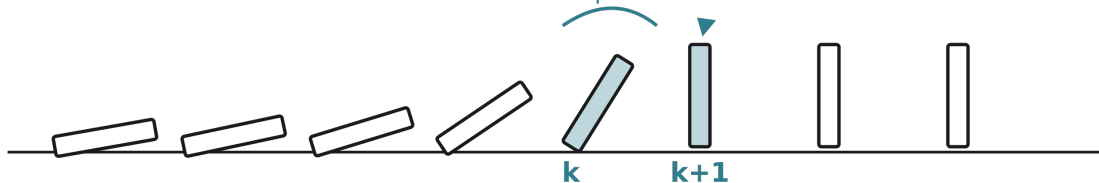
Inductive step

(2) Inductive step: $\forall k \geq a, P(k) \Rightarrow P(k+1)$

Conclusion

Then $P(n)$ is true for every integer $n \geq a$

If domino k falls, it knocks over domino $k+1$
pushes



Basis topples the first domino; the inductive step makes each falling domino topple the next.

One method, four kinds of statement

Sums

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

rewrite the sum as a closed form

Divisibility

$$3 \mid (2^{2n} - 1)$$

show $n = dk$ for some integer k

Inequalities

$$2n + 1 < 2^n$$

start the basis where it first holds

Recursive sequences

$$a_n = 2 \cdot 5^{n-1}$$

use the recurrence as the link

The same basis + inductive step proves all four - only the algebra in the step changes.

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Anatomy of an induction proof

The three moving parts

1. **Basis step:** prove the statement directly at the starting value (often $n = 1$ or $n = 0$).
2. **Inductive hypothesis:** ASSUME the statement holds for one integer $k \geq a$.
3. **Inductive step:** USING that assumption, prove the statement for $k + 1$.

A common worry, settled

"Isn't assuming $P(k)$ the same as assuming what we want?" No. We assume it for a single k and show that forces $P(k+1)$. We never assume $P(n)$ for all n . The basis plus that single link is what makes the whole chain fall.

The induction recipe - a reusable template

1. State

Write the statement $P(n)$ you want to prove, and the starting value a .

2. Basis

Check $P(a)$ directly at the smallest n - usually one line of arithmetic.

3. Hypothesis

Assume $P(k)$ holds for one arbitrary integer $k \geq a$. Write it out explicitly.

4. Step

Starting from $P(k)$, derive $P(k+1)$ by algebra. This is the real work.

5. Conclude

By induction, $P(n)$ is true for every integer $n \geq a$.

Worked proof: the coin problem

Claim (Epp, Proposition 5.2.1): every integer amount $n \geq 8$ can be paid with 3-cent and 5-cent coins.

Proof by induction on n

Basis step

$$P(8) : 8 = 3 + 5$$

Inductive step - assume some amount $k \geq 8$ is payable, and show $k + 1$ is too:

Case 1. the k -cent payment uses at least one 5 $\rightarrow$ swap one 5 for two 3s:

$$k - 5 + 6 = k + 1$$

Case 2. it uses only 3s (so at least three of them) $\rightarrow$ swap three 3s for two 5s:

$$k - 9 + 10 = k + 1$$

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Gauss and the power of a closed form

The story

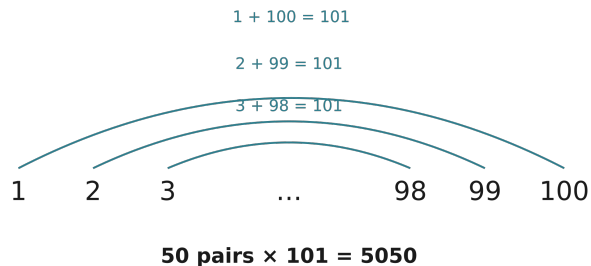
Asked to add $1 + 2 + \dots + 100$, the young Gauss paired the ends: $1 + 100$, $2 + 99$, $3 + 98$, ... Fifty pairs, each summing to 101.

$$1 + 2 + 3 + \dots + 100 = 5050$$

↑ *the sum*

↑ *the total*

A closed form gives the total directly - no loop, no adding term by term.



From a sum to a closed form

The sum (expanded form)

$$1 + 2 + 3 + \cdots + n$$

Add every term, one by one. To reach the answer you do n additions - the work grows with n .

The closed form

$$\frac{n(n+1)}{2}$$

One formula in n : multiply, add, divide. The same three operations for any n - no loop at all.

The relationship

Same number, two expressions. The sum says *what* to compute; the closed form says *how* to get it directly. They are equal for every n - and induction is exactly the proof that they always agree.

Closed forms: why we want them, and how to find one

Why a closed form is worth having

Speed one calculation, constant work - the sum needs n steps, the formula needs one.

Reach evaluate it for any n at once, even $n = 1,000,000,000$.

Insight it exposes the growth rate at a glance:

$$\frac{n(n+1)}{2} \approx \frac{n^2}{2}$$

grows like n^2 - quadratic

How we find one

- 1 Spot a pattern or use a trick - Gauss pairing, or telescoping.
- 2 Conjecture the closed form, and check it on a few small n .
- 3 Prove it holds for all n by induction - turning the guess into a theorem.

Theorem 5.2.2: sum of the first n integers

For every integer $n \geq 1$

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

↑ *the sum, written out*

↑ *the closed form*

The same thing, as a loop

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s = 0; for (k = 1; k ≤ n; k++) s = s + k;
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The loop adds the integers one at a time in n steps. The closed form $n(n+1)/2$ returns the same answer in a single step - and induction proves the two always agree.

Proof of Theorem 5.2.2, step by step

Basis (n = 1)

$$n = 1: 1 = \frac{1 \cdot 2}{2}$$

Assume P(k)

$$1 + 2 + \dots + k = \frac{k(k+1)}{2}$$

Inductive step - start at the left side and reach the goal

must reach:

$$1 + 2 + \dots + (k+1) = \frac{(k+1)(k+2)}{2}$$

start (LHS at k+1)

$$1 + 2 + \dots + k + (k+1)$$

use P(k)

$$= \frac{k(k+1)}{2} + (k+1)$$

common denominator

$$= \frac{k(k+1) + 2(k+1)}{2}$$

factor out (k+1)

$$= \frac{(k+1)(k+2)}{2}$$

Using the formula to evaluate sums

Sum of even numbers - factor the 2 out first

$$2 + 4 + \dots + 500 = 2 \cdot \frac{250 \cdot 251}{2} = 62750$$

Sum from 5 to 50 - whole sum minus the missing head

$$5 + 6 + \dots + 50 = \frac{50 \cdot 51}{2} - \frac{4 \cdot 5}{2} = 1265$$

Theorem 5.2.3: the geometric sum

For a fixed ratio $r \neq 1$ and integer $n \geq 0$

$$\sum_{i=0}^n r^i = \frac{r^{n+1} - 1}{r - 1}$$

$\uparrow$ *the geometric sum* $\uparrow$ *the closed form*

Why it matters

A geometric sum adds powers of a fixed ratio r : $1 + r + r^2 + \dots + r^n$ - compound interest, loop-doubling costs, binary place values. The closed form collapses the whole sum into one fraction.

Example (ratio $r = 2$)

$$1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$$

Proof of Theorem 5.2.3, step by step

Basis (n = 0)

$$n = 0: r^0 = 1 = \frac{r^1 - 1}{r - 1}$$

Assume P(k)

$$\sum_{i=0}^k r^i = \frac{r^{k+1} - 1}{r - 1}$$

Inductive step - split off the last term, then combine over $r - 1$

must reach:

$$\sum_{i=0}^{k+1} r^i = \frac{r^{k+2} - 1}{r - 1}$$

split off r^{k+1}

$$\sum_{i=0}^{k+1} r^i = \sum_{i=0}^k r^i + r^{k+1}$$

use P(k)

$$= \frac{r^{k+1} - 1}{r - 1} + r^{k+1}$$

common denominator

$$= \frac{r^{k+1} - 1 + r^{k+1}(r - 1)}{r - 1}$$

simplify numerator

$$= \frac{r^{k+2} - 1}{r - 1}$$

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Divisibility proof, step by step (Prop 5.3.1)

Claim: $3 \mid (2^{2n} - 1) \quad (n \geq 0)$

course notation: $d \mid n \Leftrightarrow n = dk \quad (k \in \mathbb{Z})$

Basis ($n = 0$) $n = 0 : 2^0 - 1 = 0 = 3 \cdot 0$

Assume $P(k)$ $2^{2k} - 1 = 3r$

Inductive step - peel out the previous block, then exhibit a multiple of 3

expand the power $2^{2(k+1)} - 1 = 4 \cdot 2^{2k} - 1$

peel out the block $= 4(2^{2k} - 1) + 3$

use $P(k)$: $\dots = 3r$ $= 4(3r) + 3$

factor out 3 $= 3(4r + 1)$

Inequality proof, step by step (Prop 5.3.2)

Basis ($n = 3$)

$$n = 3: 2(3) + 1 = 7 < 8 = 2^3$$

Assume $P(k)$

$$2k + 1 < 2^k$$

Inductive step - rewrite, use $P(k)$, then bound $2 \leq 2^k$ ($k \geq 3$)

must reach:

$$2(k + 1) + 1 < 2^{k+1}$$

rewrite the LHS

$$2(k + 1) + 1 = (2k + 1) + 2$$

use $P(k)$

$$< 2^k + 2$$

bound: $2 \leq 2^k$

$$\leq 2^k + 2^k$$

combine

$$= 2^{k+1}$$

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Recursive-sequence proof, step by step

The recurrence, and the closed form to prove

$$a_1 = 2, \quad a_k = 5 a_{k-1} \quad (k \geq 2) \quad \text{Claim: } a_n = 2 \cdot 5^{n-1}$$

Basis ($n = 1$)

$$n = 1: a_1 = 2 = 2 \cdot 5^0$$

Assume $P(k)$

$$a_k = 2 \cdot 5^{k-1}$$

Inductive step - use the recurrence, then $P(k)$

by the recurrence

$$a_{k+1} = 5 a_k$$

use $P(k)$

$$= 5 (2 \cdot 5^{k-1})$$

simplify

$$= 2 \cdot 5^k$$

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Inductive reasoning vs. deductive reasoning

Inductive reasoning

Starts from **observations and examples**.

Moves from **specific** → **general**.

Produces a **conjecture** - it suggests, it does not prove.

Deductive reasoning

Starts from **definitions and theorems**.

Moves from **general** → **specific**.

Produces a **proof** - the conclusion is guaranteed.

Watch the name: despite the word "induction", mathematical induction is a **deductive** method - it proves, with certainty, for all n .

Common mistakes to avoid

Skipping the basis

Without the first domino nothing falls - the step alone proves nothing.

Wrong starting value

Check where the claim first holds: $2n+1 < 2^n$ only from $n = 3$, not $n = 1$.

Misreading the hypothesis

You assume $P(k)$ for one k , not $P(n)$ for all n . That is not circular.

Off-by-one in the step

Be sure what the $(k+1)$ -th term actually is before adding it on.

Stopping early

Reaching the closed form is not enough - state the conclusion for all n .

More statements you can now prove by induction

Each follows the same recipe - basis, hypothesis, step. Sums and divisibility and inequalities, all in one toolkit:

Sum formula	$\sum_{i=1}^n i(i+1) = \frac{n(n+1)(n+2)}{3}$
Sum of cubes	$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$
Divisibility	$5 \mid (8^n - 3^n)$
Divisibility	$3 \mid (n^3 + 2n)$
Inequality	$n^2 < 2^n \quad (n \geq 5)$

Recommended exercises (Epp, 4th ed.)

5.2 Model a problem like the coins / postage	1, 2
5.2 Prove a sum formula directly	6, 7, 8, 9
5.2 Prove the standard formulas ($\sum i^2$, $\sum i^3$, ...)	10, 11, 12, 13, 17
5.2 Evaluate a sum using the theorems	20, 21, 22, 23, 25, 26
5.3 Divisibility by induction	8, 9, 10, 11, 12, 14, 15
5.3 Inequalities by induction	16, 17, 18, 19, 20, 23
5.3 Closed form of a recursive sequence	24, 25, 26, 27

Summary

- Mathematical induction proves a statement for all integers $n \geq a$ using two steps: a basis case, and an inductive step $P(k) \Rightarrow P(k+1)$.
- The inductive hypothesis is the assumption $P(k)$ for one k ; you use it to force $P(k+1)$. It is not circular.
- Sum of the first n integers: $n(n+1)/2$. Geometric sum: $(r^{n+1} - 1)/(r - 1)$.
- The same method proves divisibility ($d \mid n$ means $n = d k$, so exhibit the multiple) and inequalities (start the basis where the claim first holds).
- For a recursive sequence, the recurrence supplies the link the inductive step needs to prove a closed form.
- Despite its name, mathematical induction is a deductive method - it proves with certainty, for every n .